

GAMBLING TASK · OPAL04 · D1/D2 MODULATION

Replicating Sasaki et al. (2024)

Single trained network — `gambling_d1d2_plasticity_pos_reg_opal04`. Generated 2026-05-30 19:25.

Riskiness (DA=0)

0.54

P(risky) at neutral dopamine

Modulation span

0.88

Δ riskiness, ctx $-0.9 \rightarrow +0.9$

Context levels

19

pre-computed.pkl files

Figures

26

panels generated

Figure index

SECTION	FIGURE	DESCRIPTION
Pre-flight	Learning curve	Mean reward vs training iteration.
Pre-flight	Critic health	Value-network predictions vs Monte-Carlo returns.

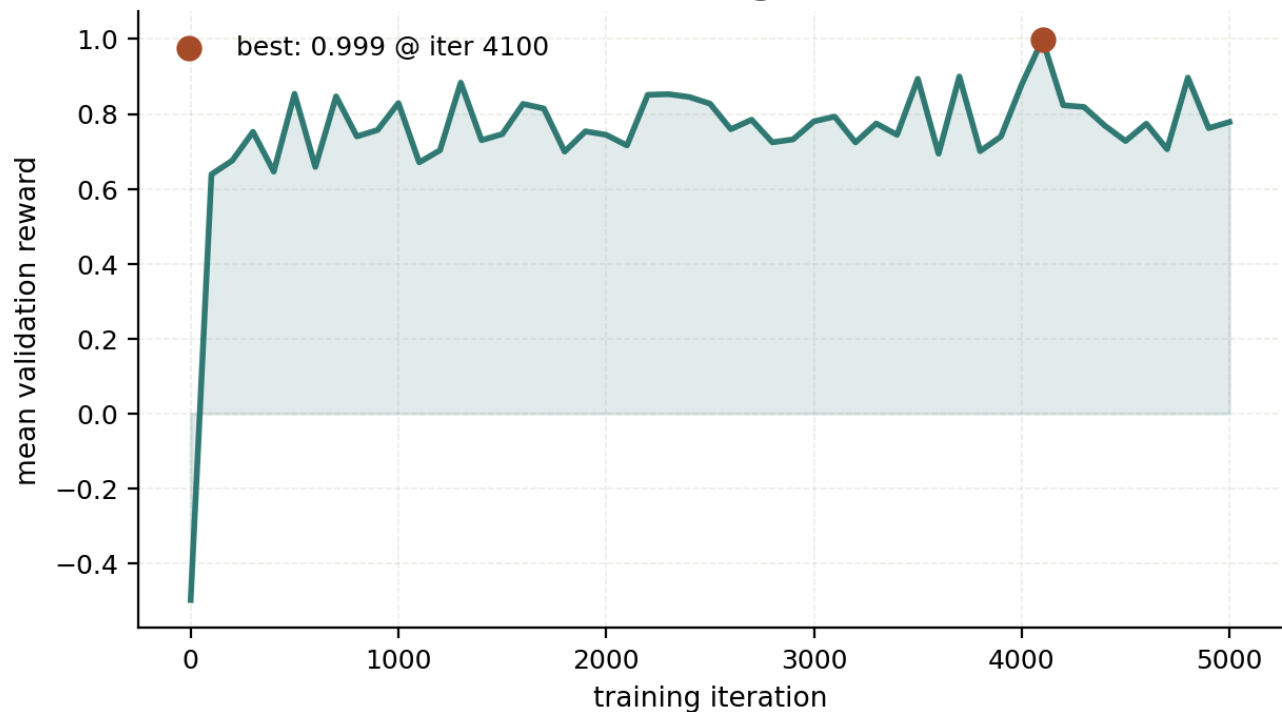
SECTION	FIGURE	DESCRIPTION
Baseline behaviour	Dose-response summary (riskiness · EV accuracy · completion)	Riskiness, EV accuracy, and completion rate across all dopamine context levels.
Baseline behaviour	Choice frequency heatmap with marginals	Transposed 5×5 heatmap (x=probability, y=EV). Top marginal = risk psychometric with 95% bootstrap CI
Baseline behaviour	Prospect-theory utility curve	Monkeys showed convex utility (risk-seeking) and/or inverse-S probability weighting (overweighting o
Baseline behaviour	Prospect-theory utility curves across all DA levels	Utility curvature (α) and probability weighting (γ) across all ctx levels.
Baseline behaviour	Psychometrics at 5 dopamine levels	Higher DA → steeper risk-seeking slope; lower DA → risk-averse.
Baseline behaviour	Probability curves — all ctx levels (pre-computed)	Psychometric curves at every context level.
Baseline behaviour	Proportion chosen by probability (marginal risk slope)	Proportion chosen by probability (marginal risk slope)
Baseline behaviour	Proportion chosen by expected value (marginal EV slope)	Proportion chosen by expected value (marginal EV slope)
Necessity — ablation	Ablation: Δ slope dot plot (risk-axis vs value-ctrl)	Δ risk slope: risk-axis=-0.014, value-ctrl=+0.145. Δ EV slope: risk-axis=+nan.
Directional DA control	Riskiness dose-response (model diagnostic)	Riskiness as a function of dopamine context level. DA-risk correlation = +0.97, span = 0.88, neutral
Directional DA control	Risk preference across ctx levels (pre-computed sigmoid)	Risk preference across ctx levels (pre-computed sigmoid)
Directional DA control	EV preference across ctx levels (pre-computed)	EV preference across ctx levels (pre-computed)
Directional DA control	Directional stimulation psychometrics (Δ prob + Δ EV)	Δ probability psychometric curves for D1/D2-only stimulation, with cumulative Gaussian fit and 95% bo
Directional DA control	Risk threshold ON vs OFF — bootstrap cloud	Bootstrap pseudo-sessions: D1 stim cloud above diagonal, D2 cloud below, control on it. D1 Δ =-0.00,

SECTION	FIGURE	DESCRIPTION
Directional DA control	EV sensitivity vs dopamine (dissociation)	EV slope across all ctx levels. std=0.066 (small = EV sensitivity unchanged by dopamine).
Timing specificity	Phase-restricted modulation — sliding window + psychometrics	Sliding-window sweep shows when context has the biggest effect. Epoch bars + psychometric overlays c
Neural decoding	Decoder psychometric + risk axis trajectory	Left: decoder P(risky) vs offered probability with bootstrap CI. Right: mean ± SEM projection onto r
Neural decoding	Riskiness ON vs OFF scatter (bootstrap)	Bootstrap pseudo-sessions: riskiness stim-OFF (x) vs D1-stim ON (y). OFF=0.54, ON=0.81, Δ=+0.27.
Neural representations	Neural regression β scatter	Neurons encoding EV only, risk only, or mixed — D1/D2 split.
Neural representations	vs D2 pathway pull across ctx	D1 pull grows with DA; D2 shrinks — push-pull mechanism.
Neural representations	Existing: V and RPE dynamics across all ctx levels	Value prediction and objective RPE traces for 10 representative trials at each context level.
Neural representations	Existing: Policy subjective values mega (5 rows × 19 ctx)	D1 pull, D2 pull, total logit, engagement drive, and choice advantage across all 19 context levels.
Neural representations	Existing: Mega comparison context sweep	Comprehensive multi-row comparison of behaviour, neural regression, and value grids across all dopam

Pre-flight

Learning curve

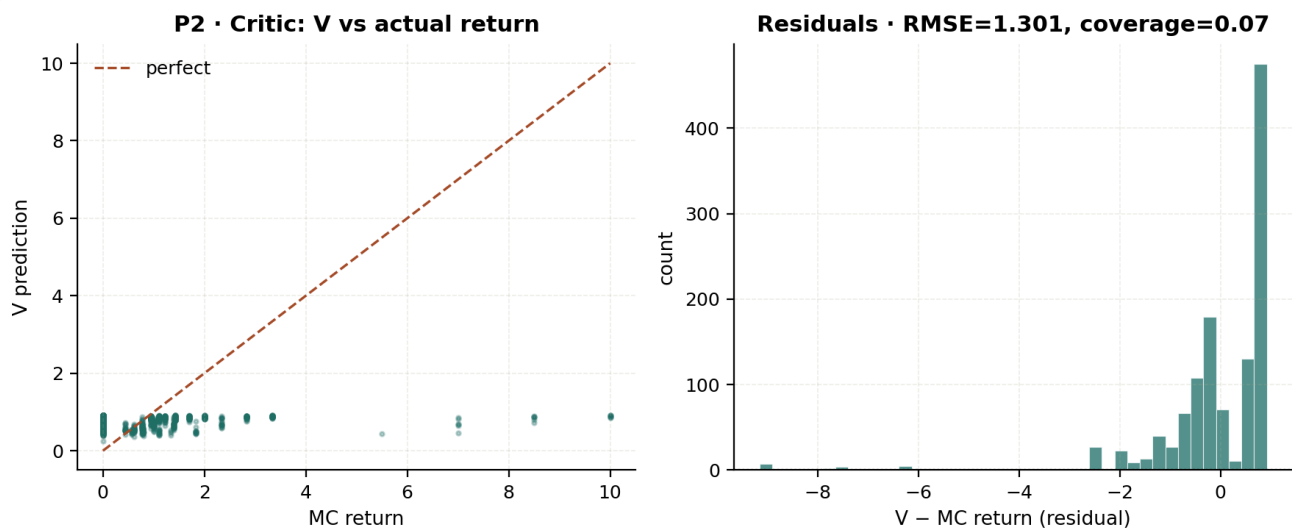
Mean reward vs training iteration.

P1 · Learning curve

Mean reward vs training iteration.

Critic health

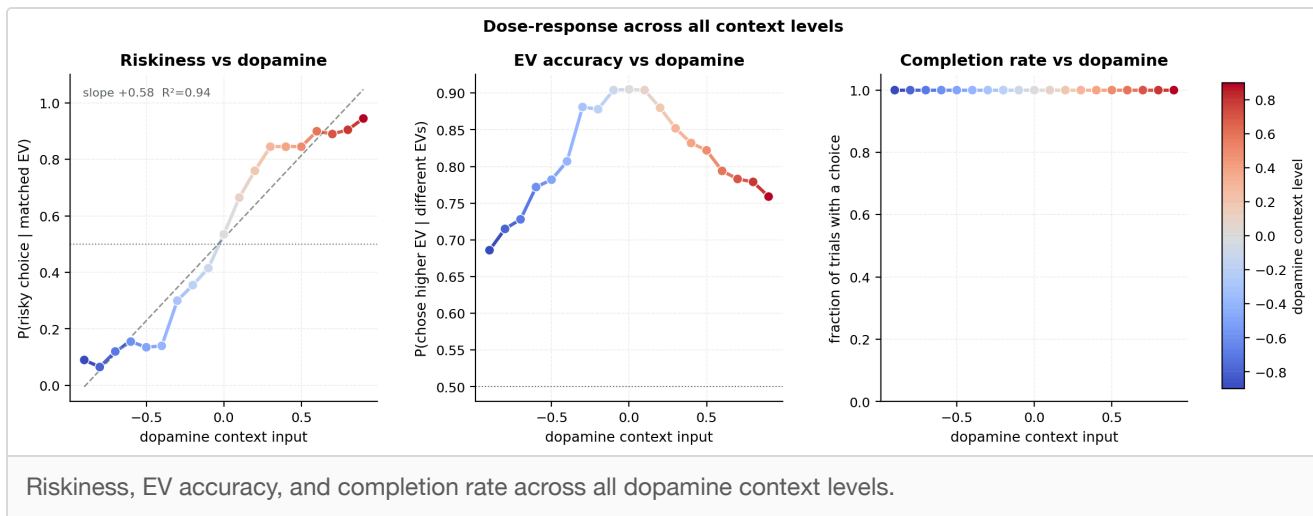
Value-network predictions vs Monte-Carlo returns.



Value-network predictions vs Monte-Carlo returns.

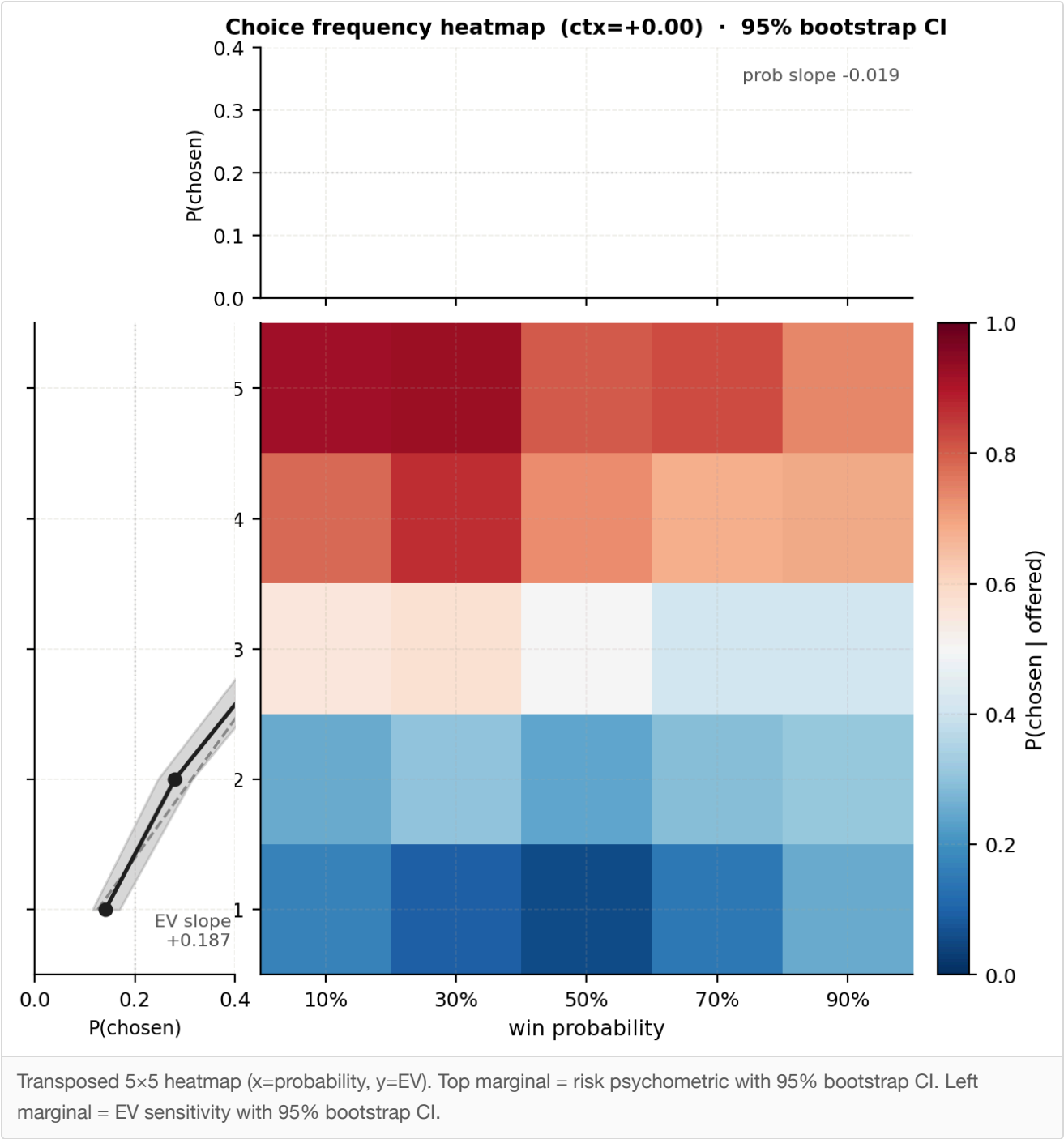
Baseline behaviour**Dose-response summary (riskiness · EV accuracy · completion)**

Riskiness, EV accuracy, and completion rate across all dopamine context levels.



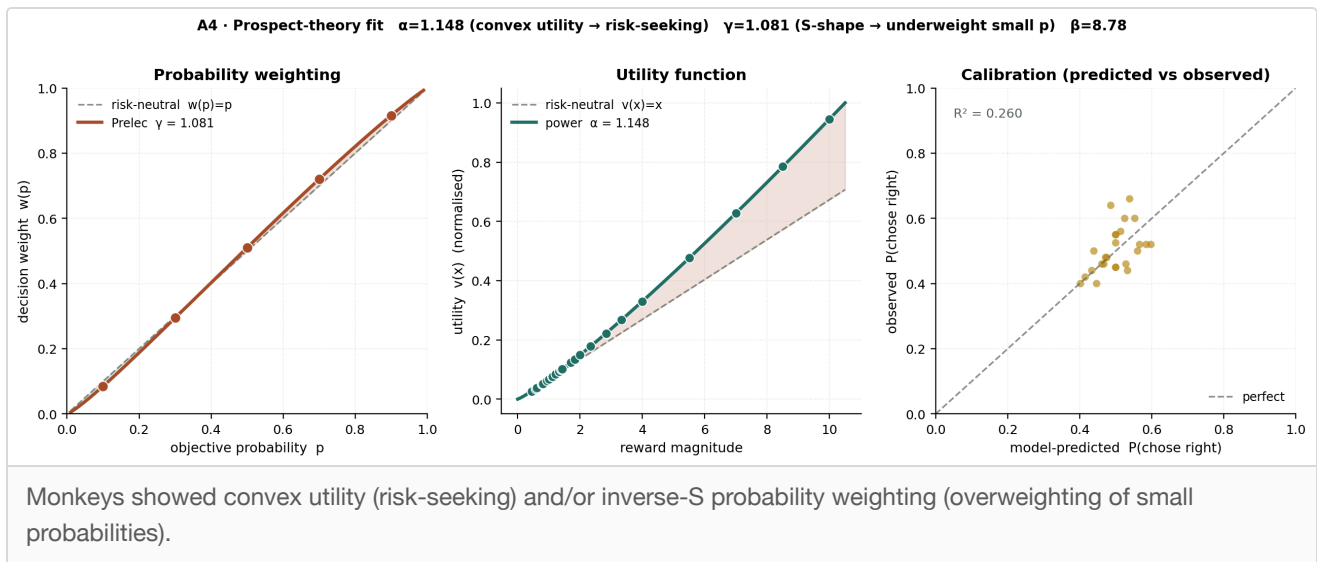
Choice frequency heatmap with marginals

Transposed 5x5 heatmap (x=probability, y=EV). Top marginal = risk psychometric with 95% bootstrap CI. Left marginal = EV sensitivity with 95% bootstrap CI.



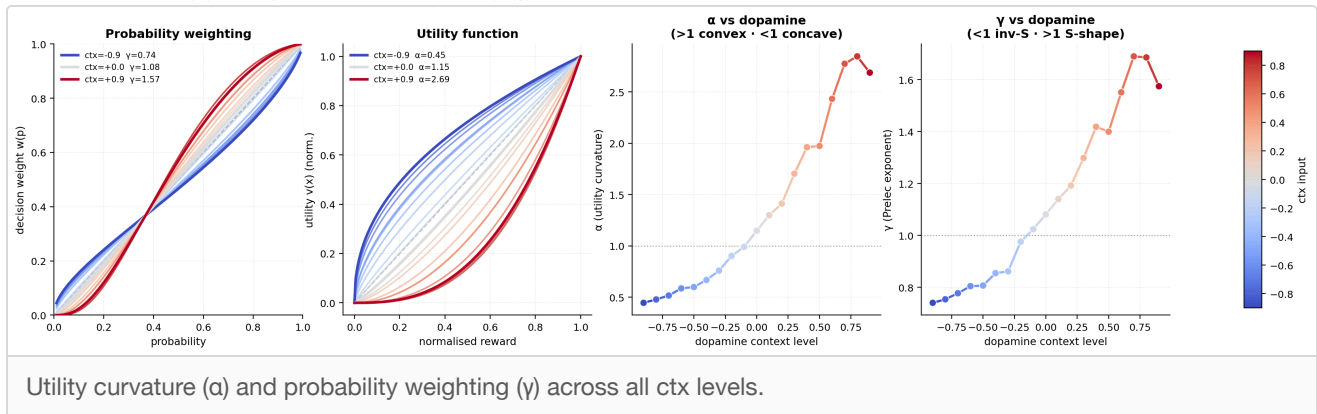
Prospect-theory utility curve

Monkeys showed convex utility (risk-seeking) and/or inverse-S probability weighting (overweighting of small probabilities).



Prospect-theory utility curves across all DA levels

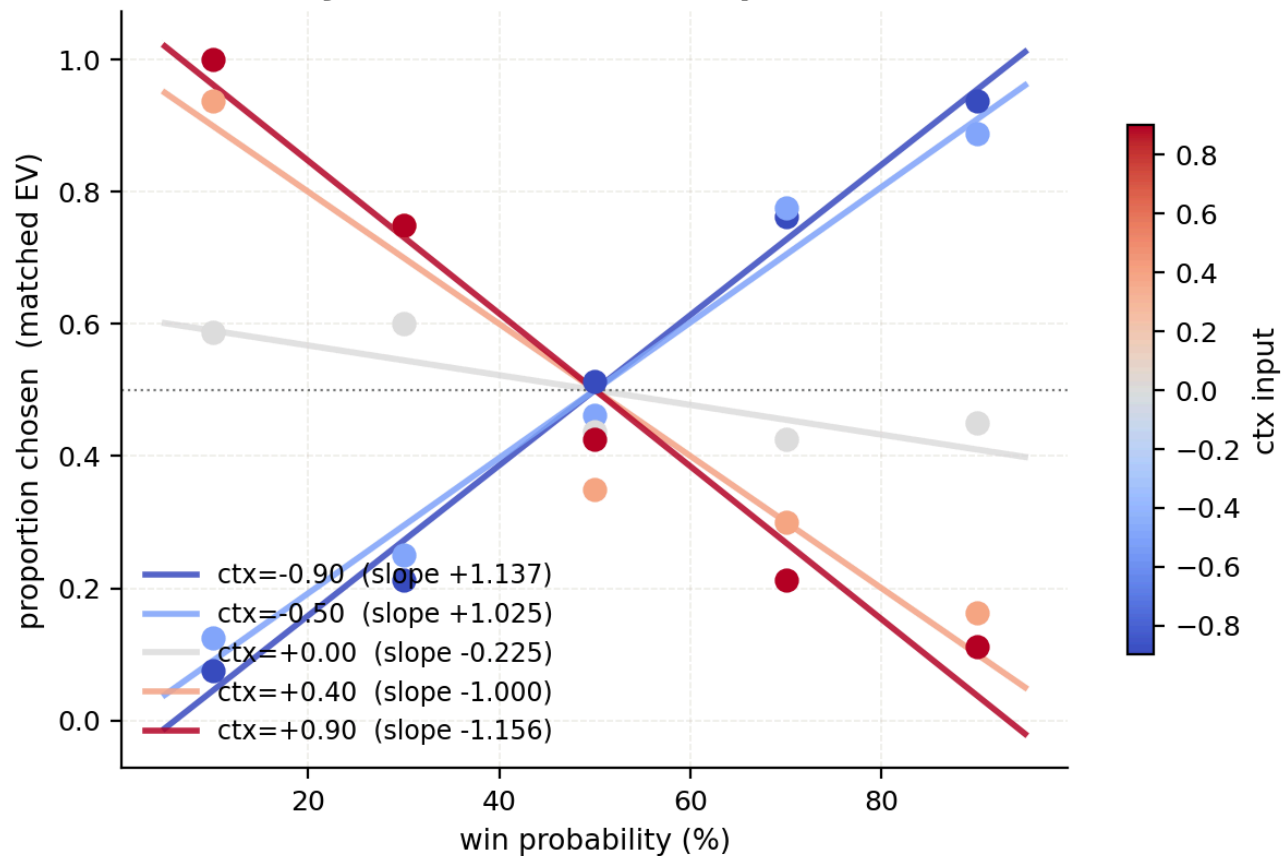
Utility curvature (α) and probability weighting (γ) across all ctx levels.



Psychometrics at 5 dopamine levels

Higher DA → steeper risk-seeking slope; lower DA → risk-averse.

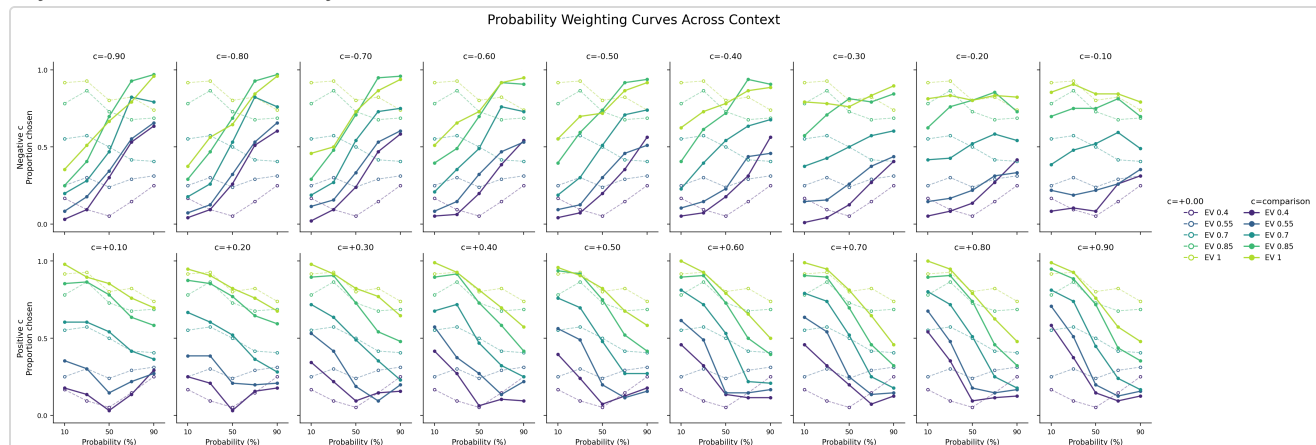
A4 · Psychometrics across dopamine levels



Higher DA → steeper risk-seeking slope; lower DA → risk-averse.

Probability curves — all ctx levels (pre-computed)

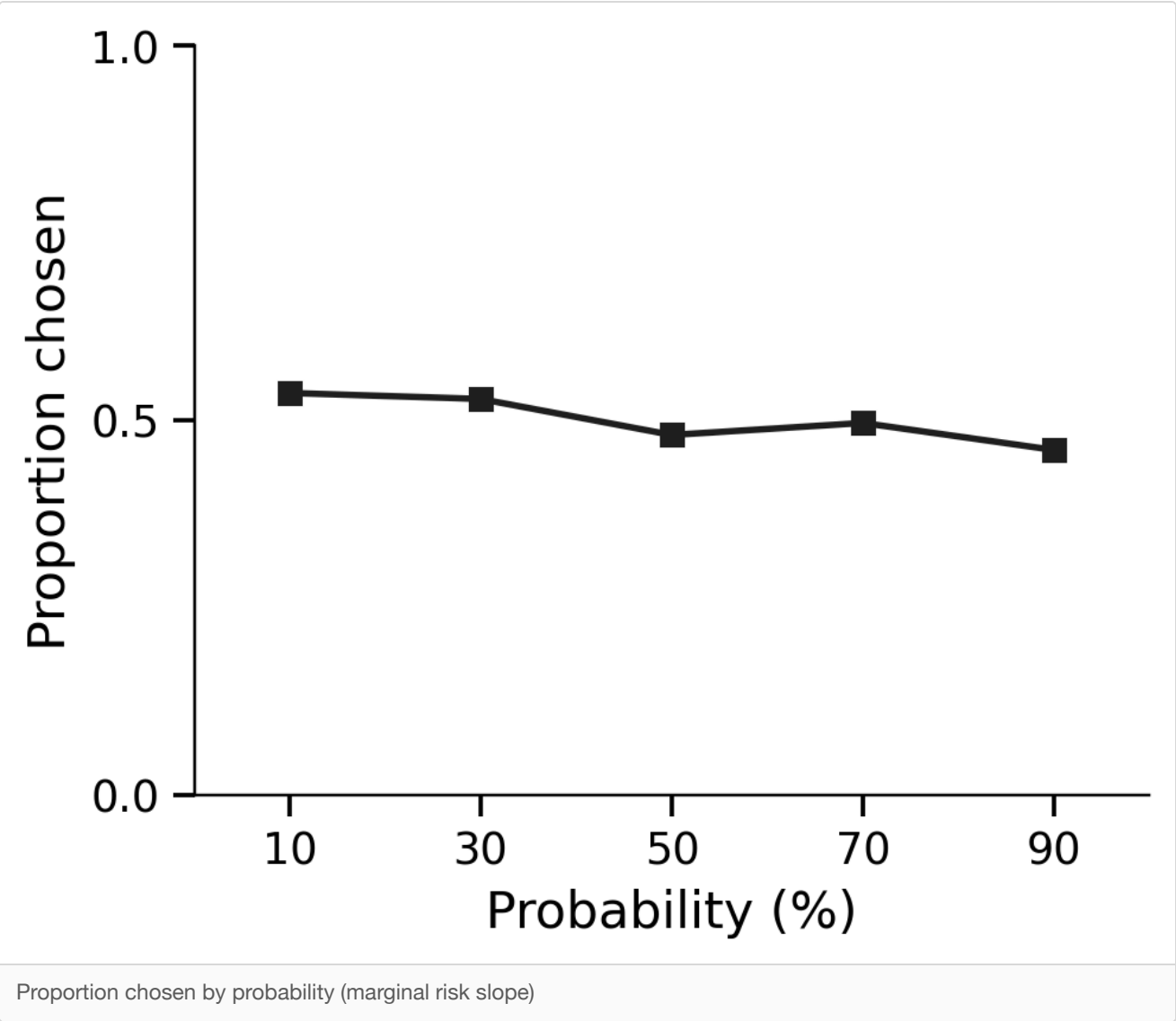
Psychometric curves at every context level.



Psychometric curves at every context level.

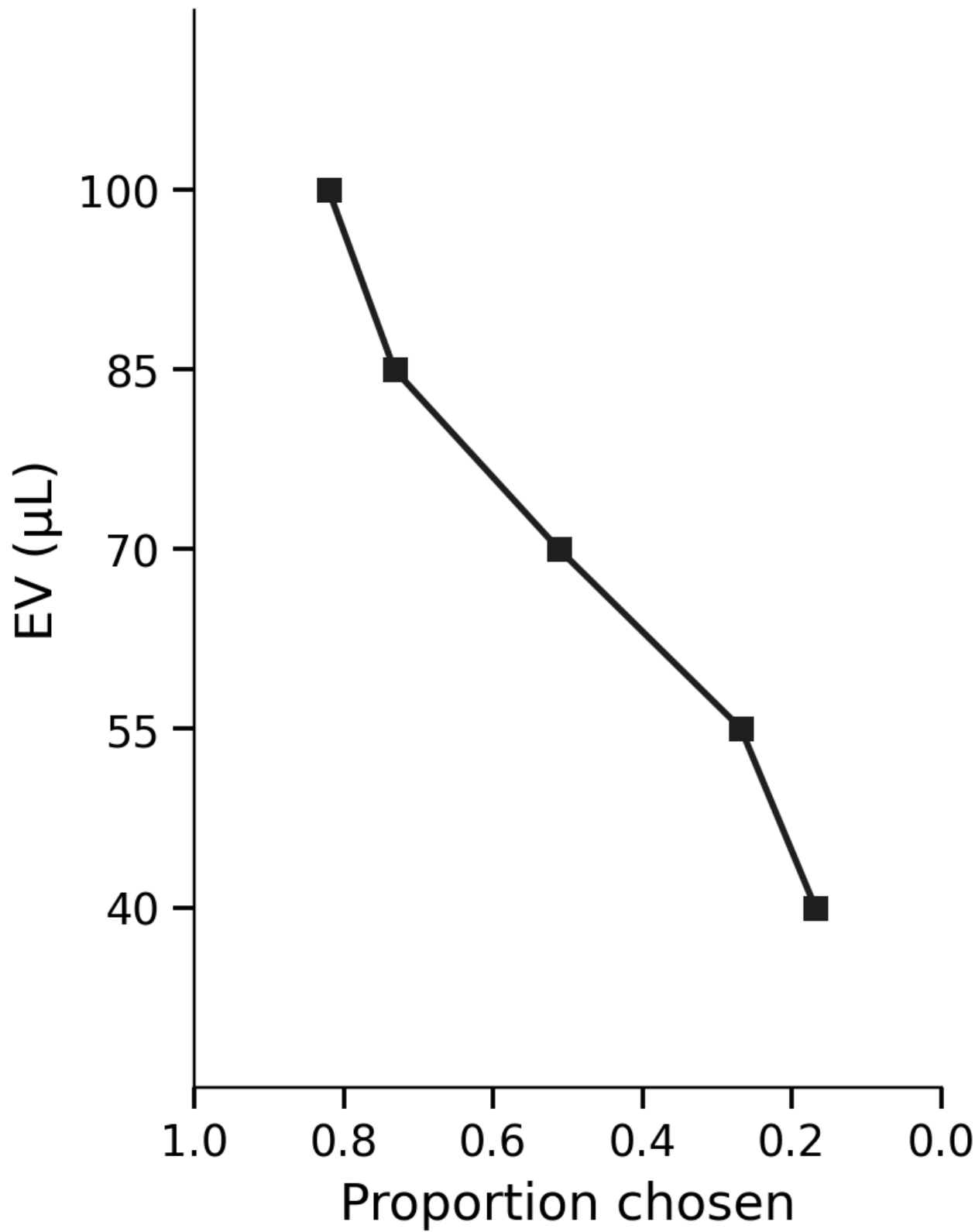
Proportion chosen by probability (marginal risk slope)

Proportion chosen by probability (marginal risk slope)



Proportion chosen by expected value (marginal EV slope)

Proportion chosen by expected value (marginal EV slope)

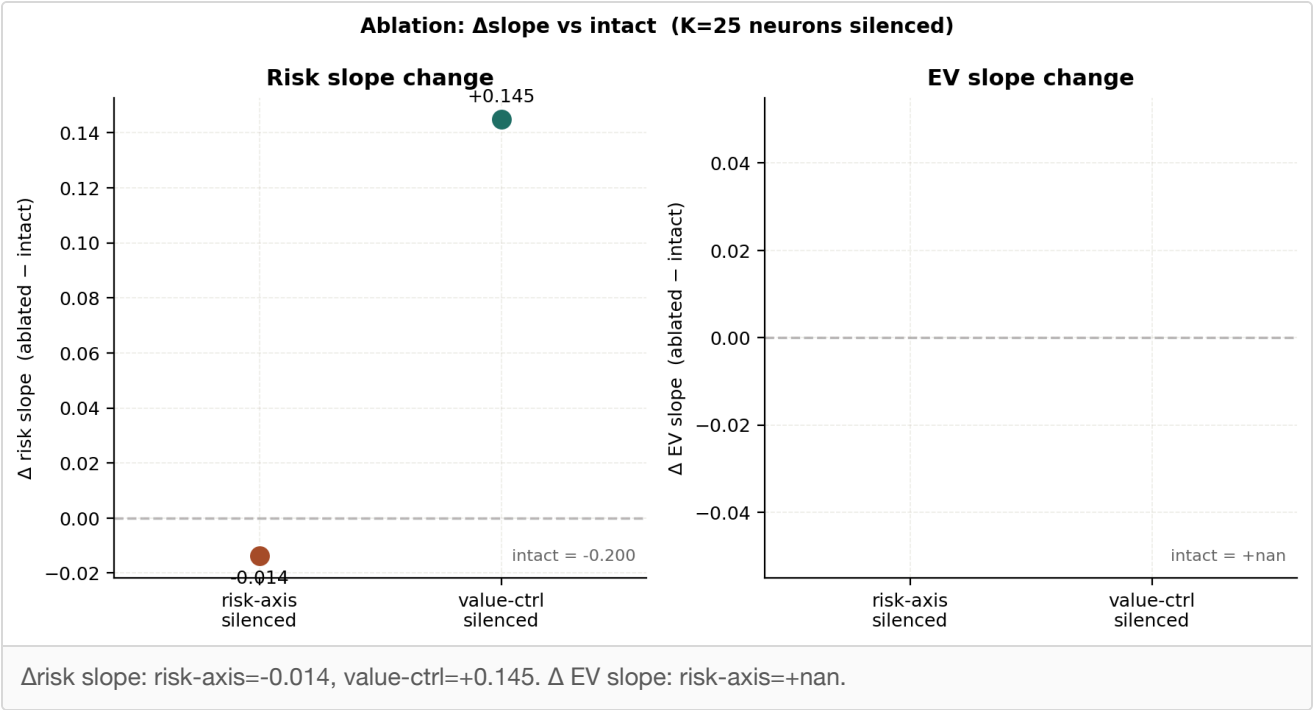


Proportion chosen by expected value (marginal EV slope)

Necessity — ablation

Ablation: Δslope dot plot (risk-axis vs value-ctrl)

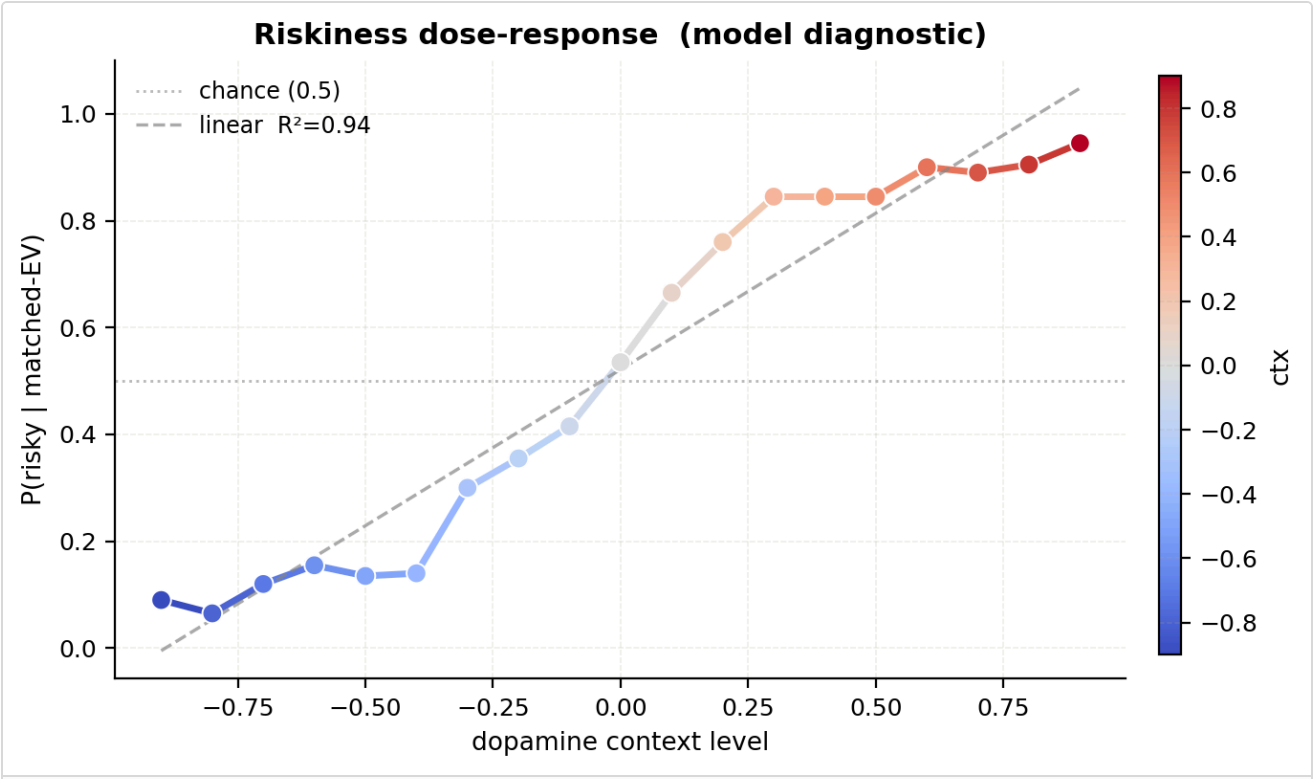
Δ risk slope: risk-axis=-0.014, value-ctrl=+0.145. Δ EV slope: risk-axis=+nan.



Directional DA control

Riskiness dose-response (model diagnostic)

Riskiness as a function of dopamine context level. DA-risk correlation = +0.97, span = 0.88, neutral riskiness = 0.54
→ MONOTONIC ✓



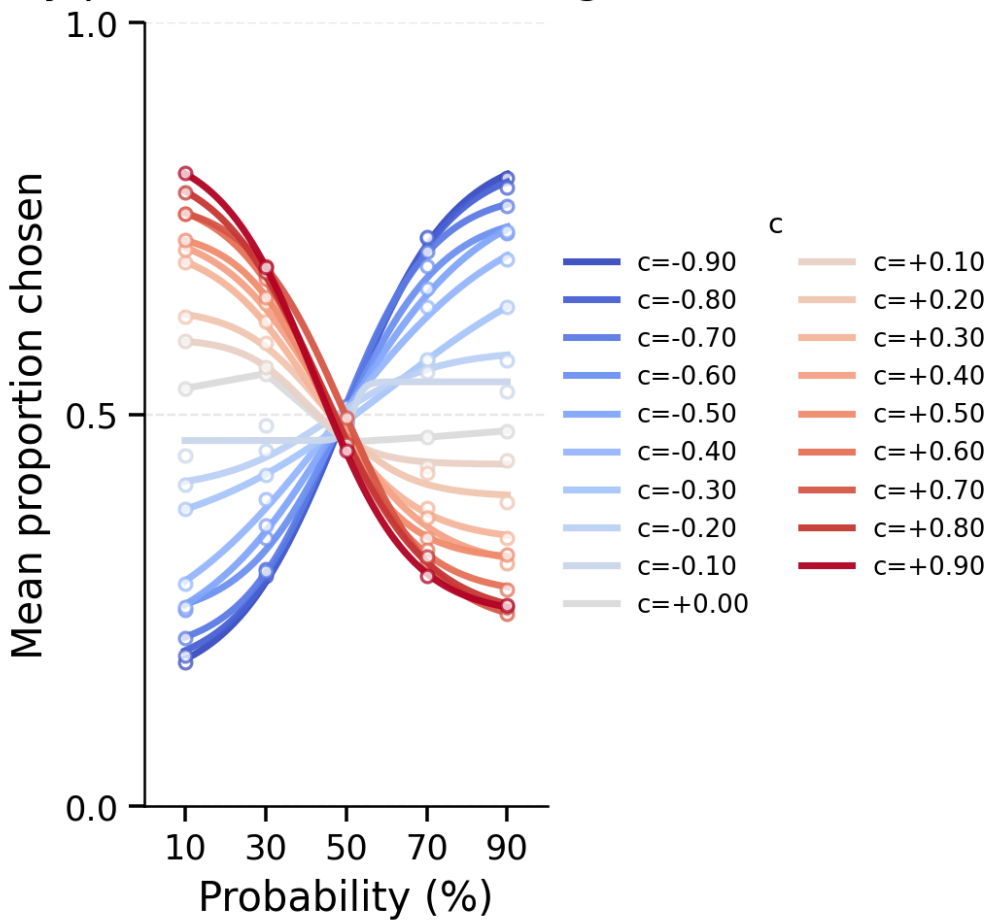
Riskiness as a function of dopamine context level. DA-risk correlation = +0.97, span = 0.88, neutral riskiness = 0.54

→ MONOTONIC ✓

Risk preference across ctx levels (pre-computed sigmoid)

Risk preference across ctx levels (pre-computed sigmoid)

Risk/probability preference across c (sigmoid-smoothed)

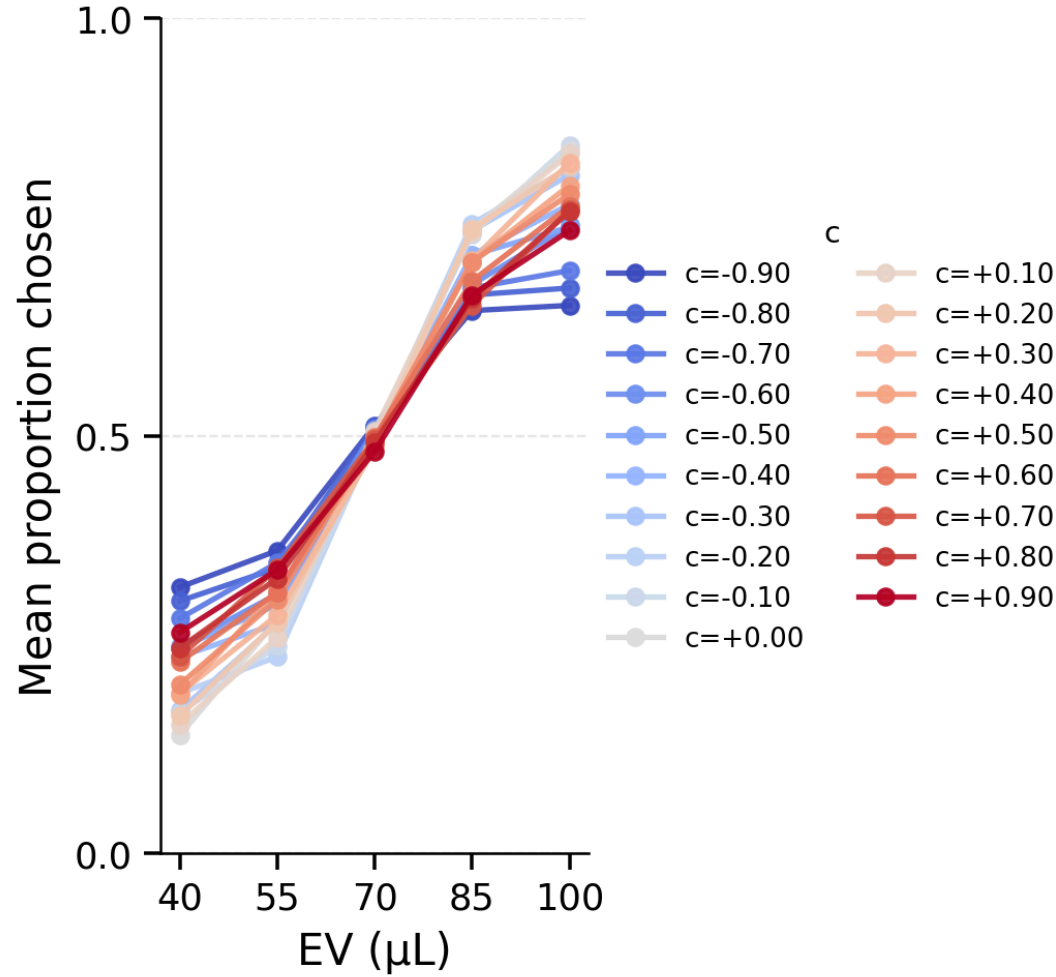


Risk preference across ctx levels (pre-computed sigmoid)

EV preference across ctx levels (pre-computed)

EV preference across ctx levels (pre-computed)

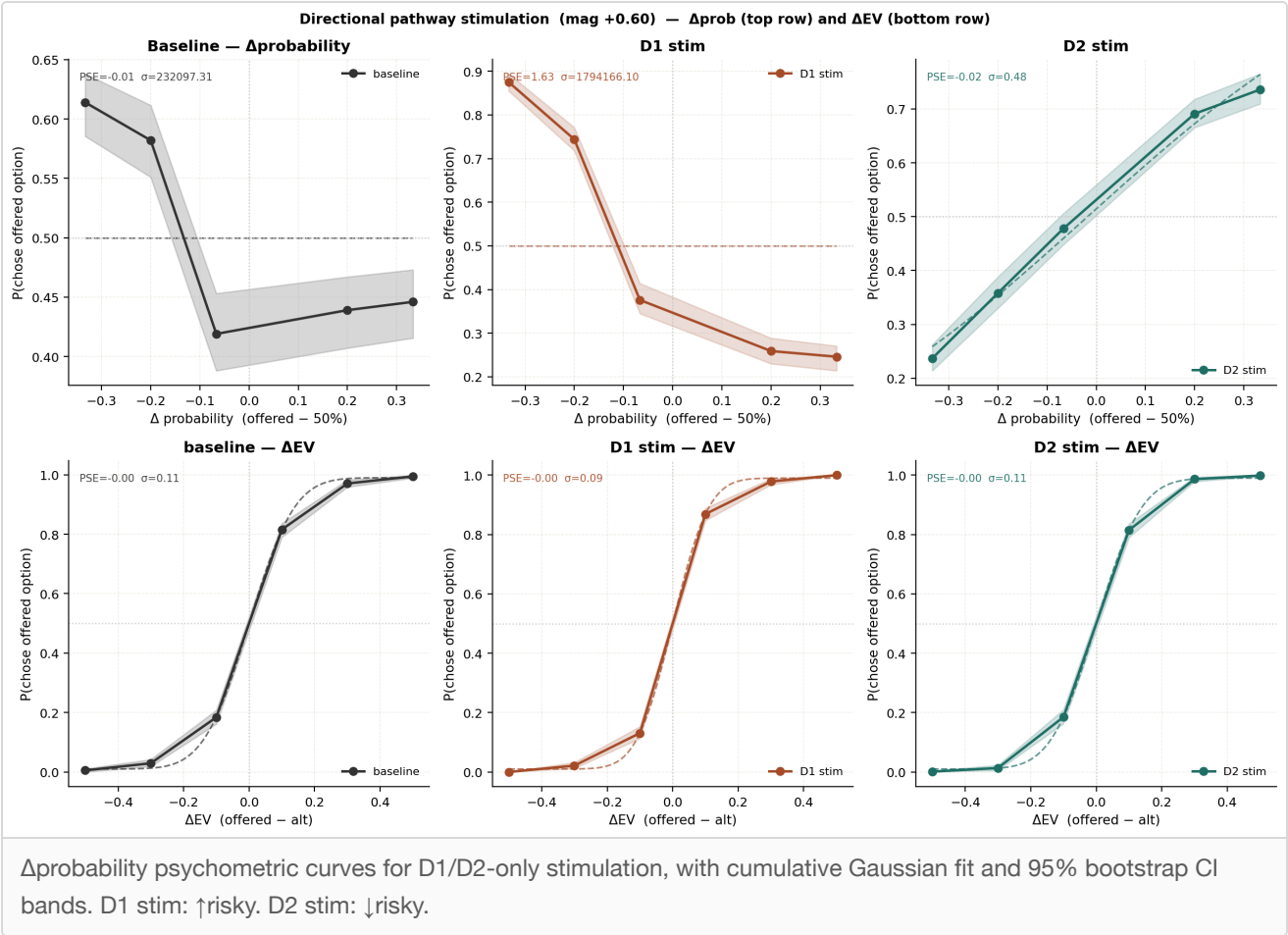
EV preference across c (direct point connections)



EV preference across ctx levels (pre-computed)

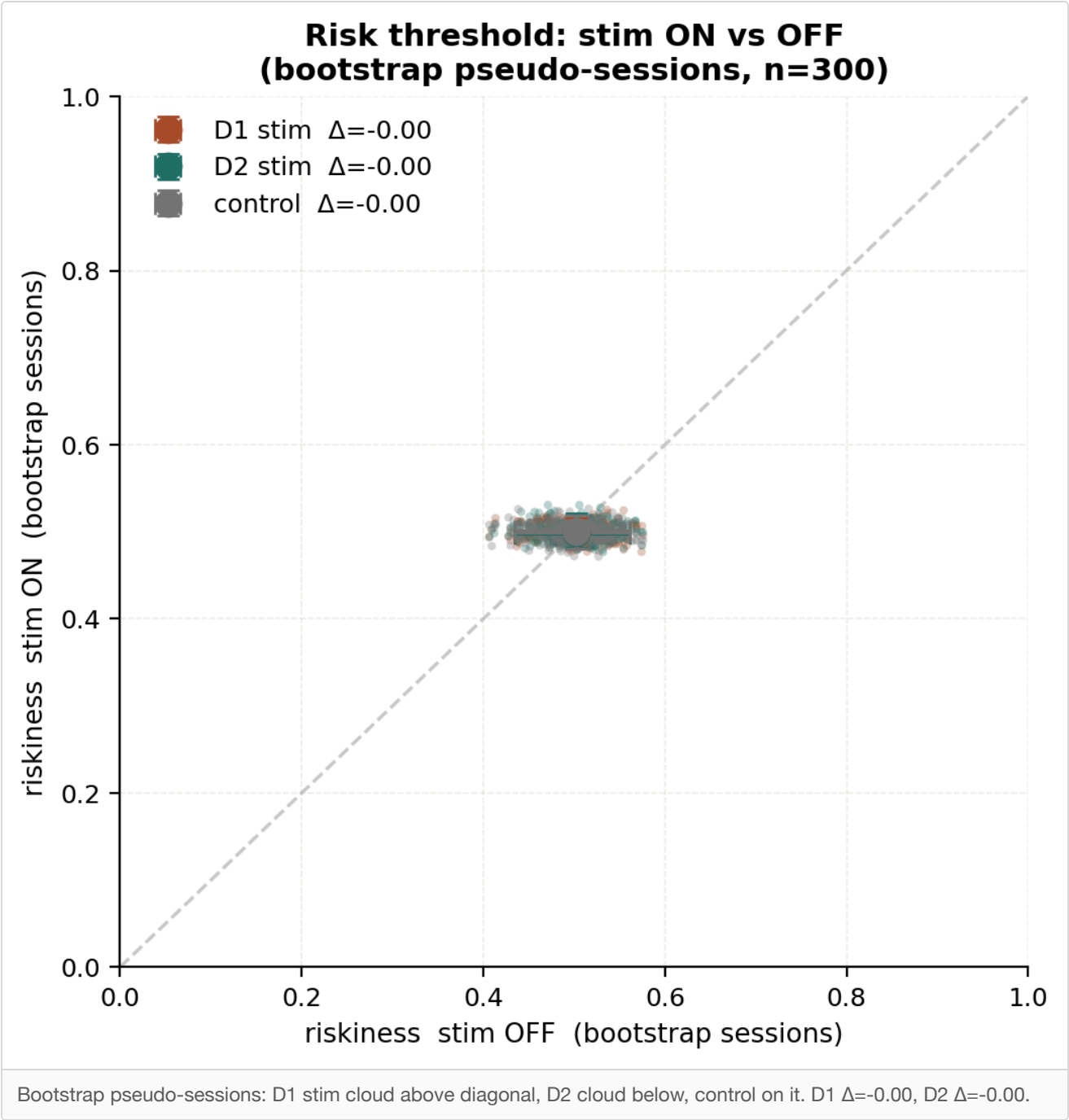
Directional stimulation psychometrics ($\Delta\text{prob} + \Delta\text{EV}$)

Δ probability psychometric curves for D1/D2-only stimulation, with cumulative Gaussian fit and 95% bootstrap CI bands. D1 stim: $\uparrow$ risky. D2 stim: $\downarrow$ risky.



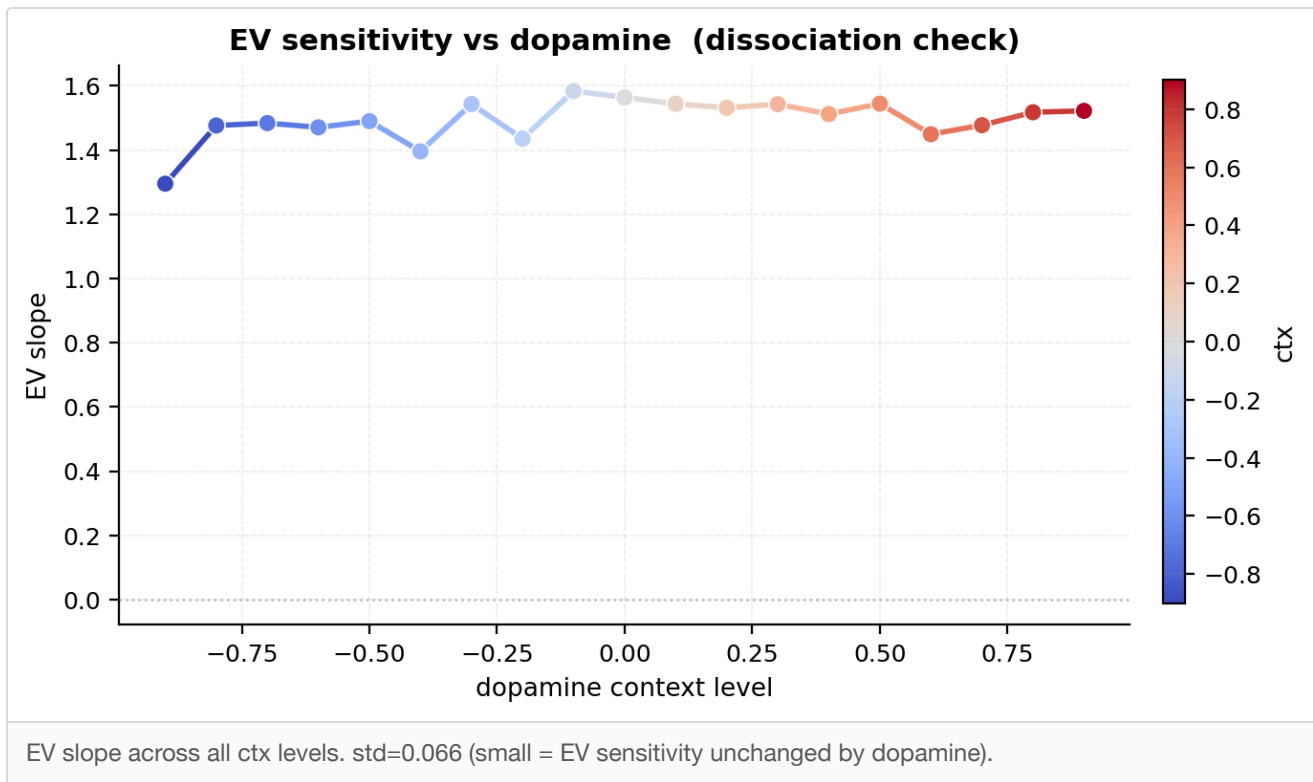
Risk threshold ON vs OFF — bootstrap cloud

Bootstrap pseudo-sessions: D1 stim cloud above diagonal, D2 cloud below, control on it. D1 Δ=-0.00, D2 Δ=-0.00.



EV sensitivity vs dopamine (dissociation)

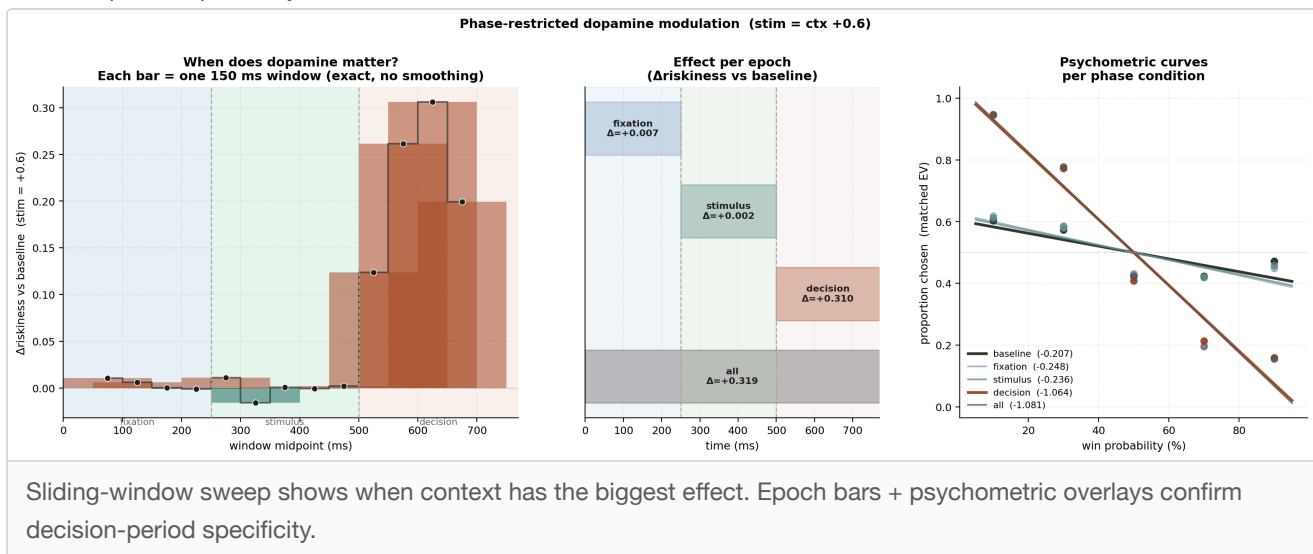
EV slope across all ctx levels. std=0.066 (small = EV sensitivity unchanged by dopamine).



Timing specificity

Phase-restricted modulation — sliding window + psychometrics

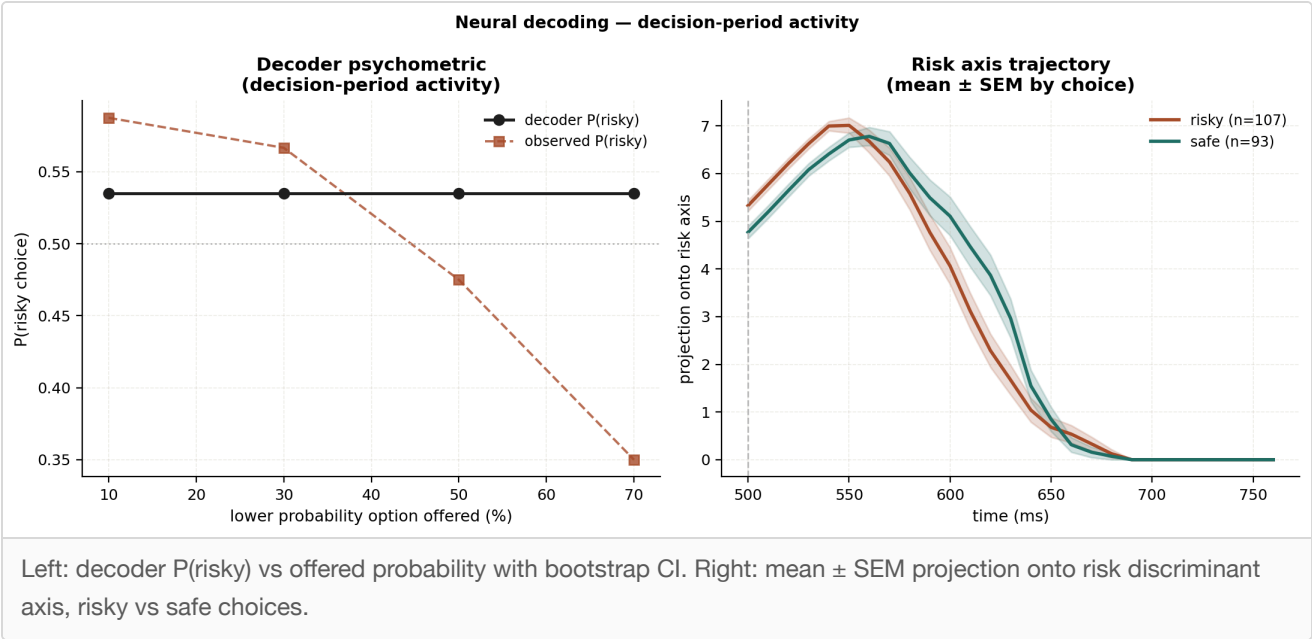
Sliding-window sweep shows when context has the biggest effect. Epoch bars + psychometric overlays confirm decision-period specificity.



Neural decoding

Decoder psychometric + risk axis trajectory

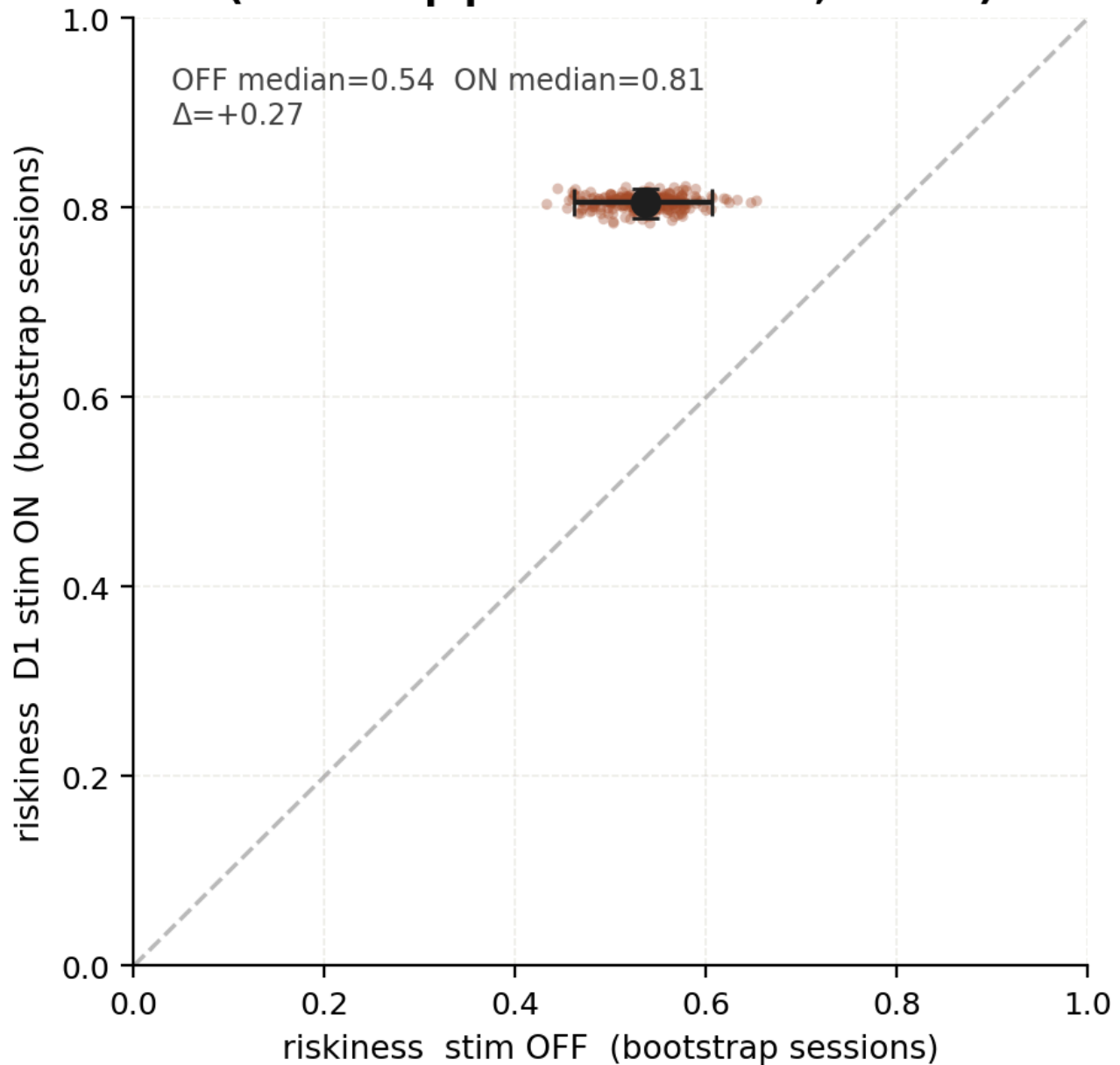
Left: decoder P(risky) vs offered probability with bootstrap CI. Right: mean ± SEM projection onto risk discriminant axis, risky vs safe choices.



Riskiness ON vs OFF scatter (bootstrap)

Bootstrap pseudo-sessions: riskiness stim-OFF (x) vs D1-stim ON (y). OFF=0.54, ON=0.81, Δ=+0.27.

Decoder shift: riskiness ON vs OFF (bootstrap pseudo-sessions, n=250)

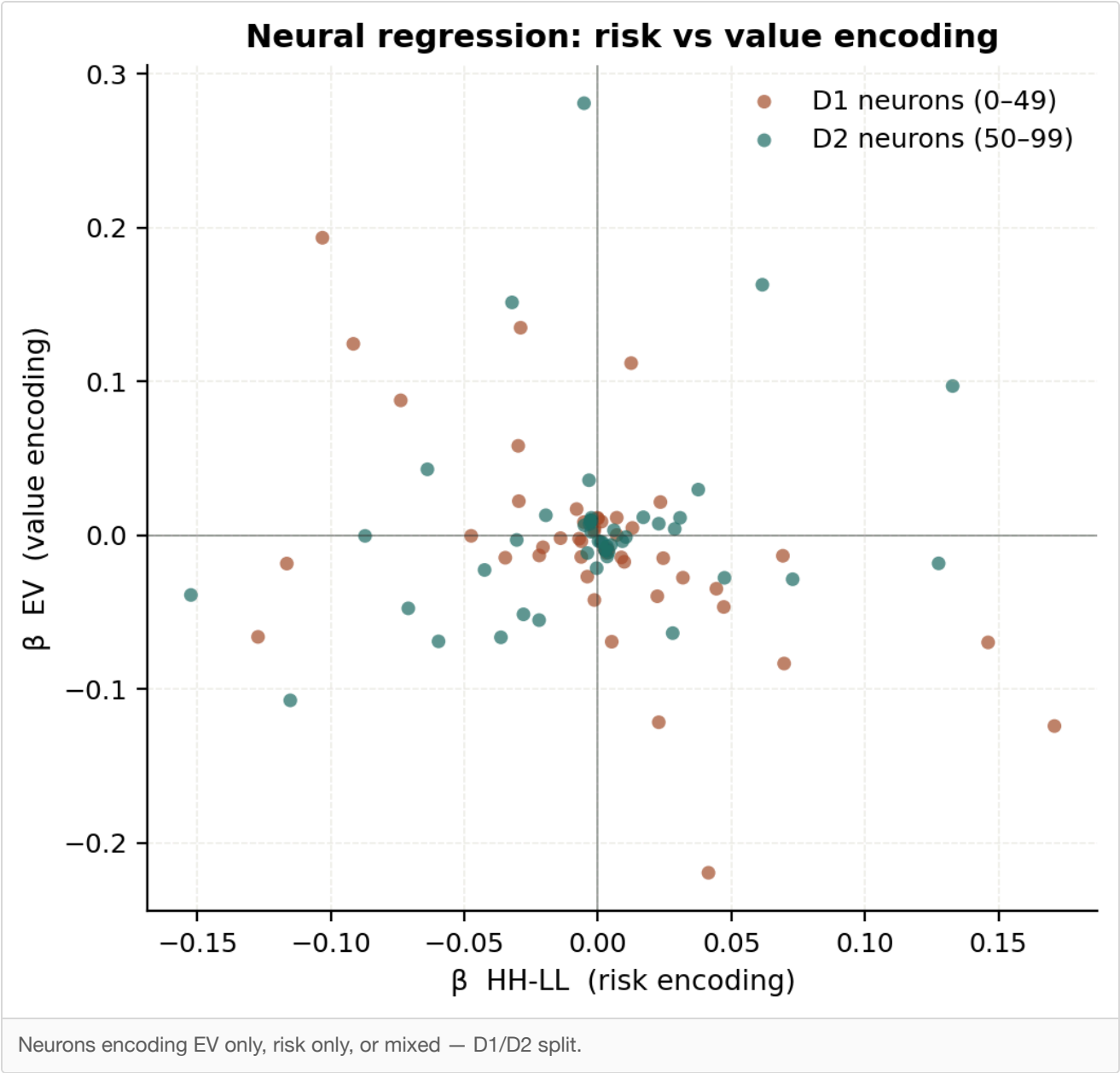


Bootstrap pseudo-sessions: riskiness stim-OFF (x) vs D1-stim ON (y). OFF=0.54, ON=0.81, $\Delta=+0.27$.

Neural representations

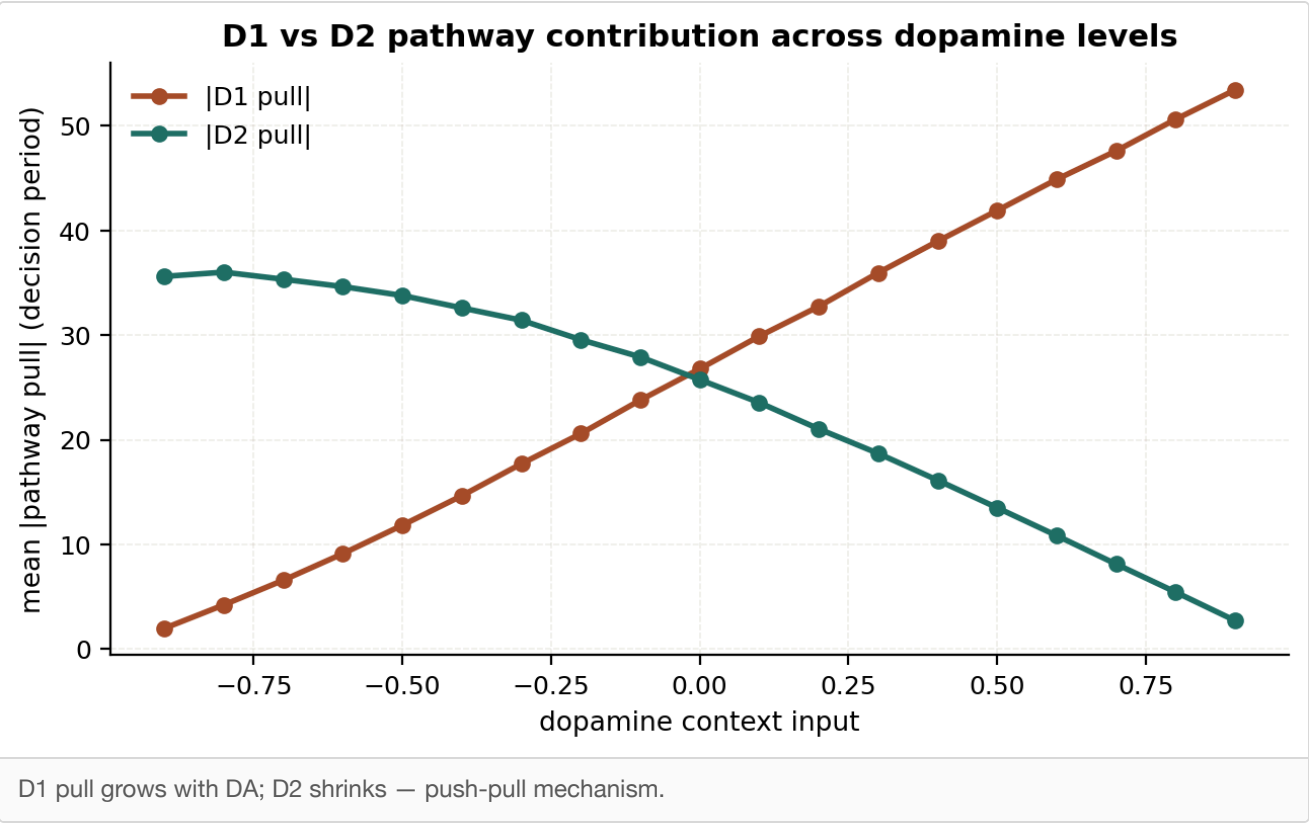
Neural regression β scatter

Neurons encoding EV only, risk only, or mixed — D1/D2 split.



vs D2 pathway pull across ctx

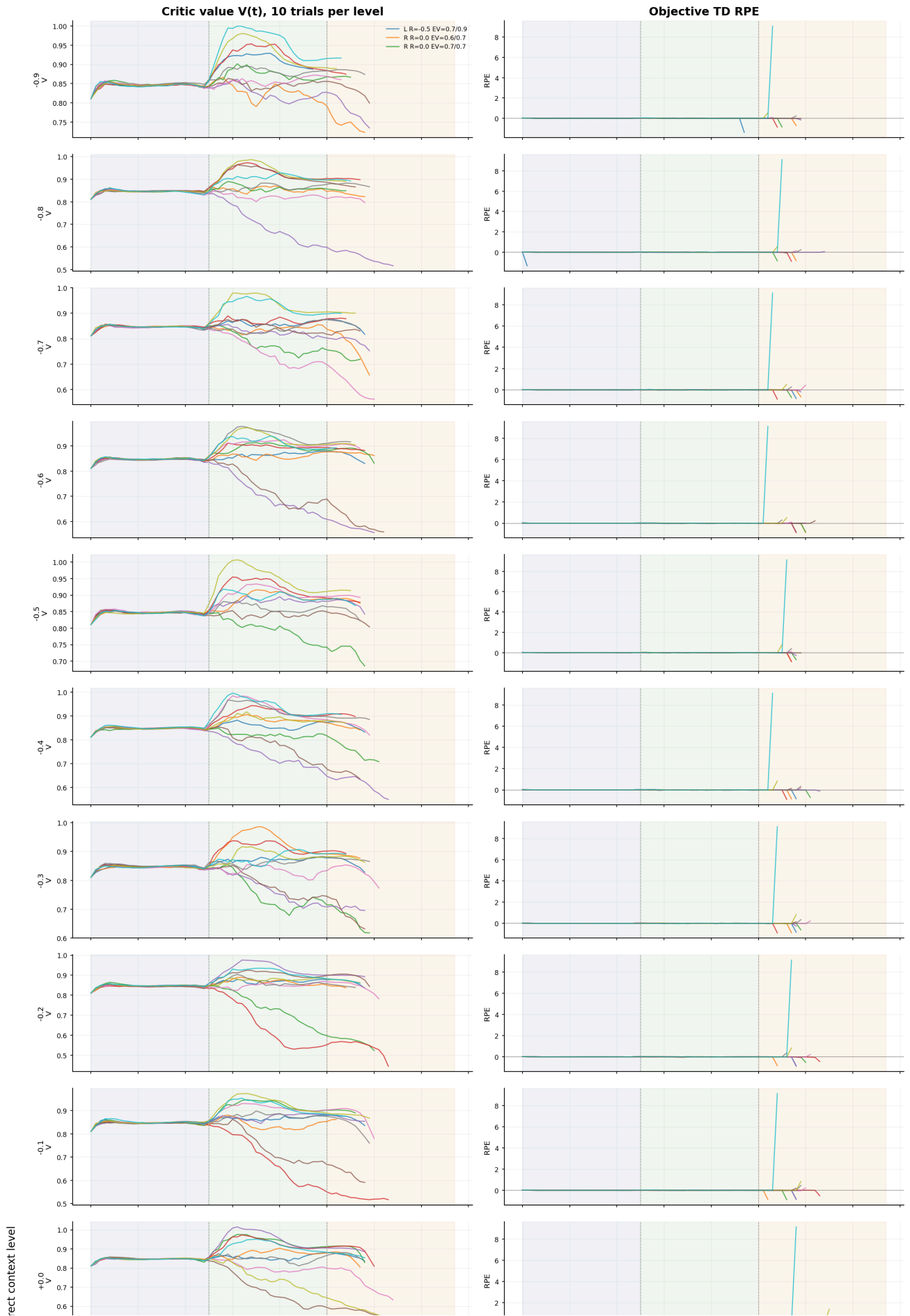
D1 pull grows with DA; D2 shrinks — push-pull mechanism.



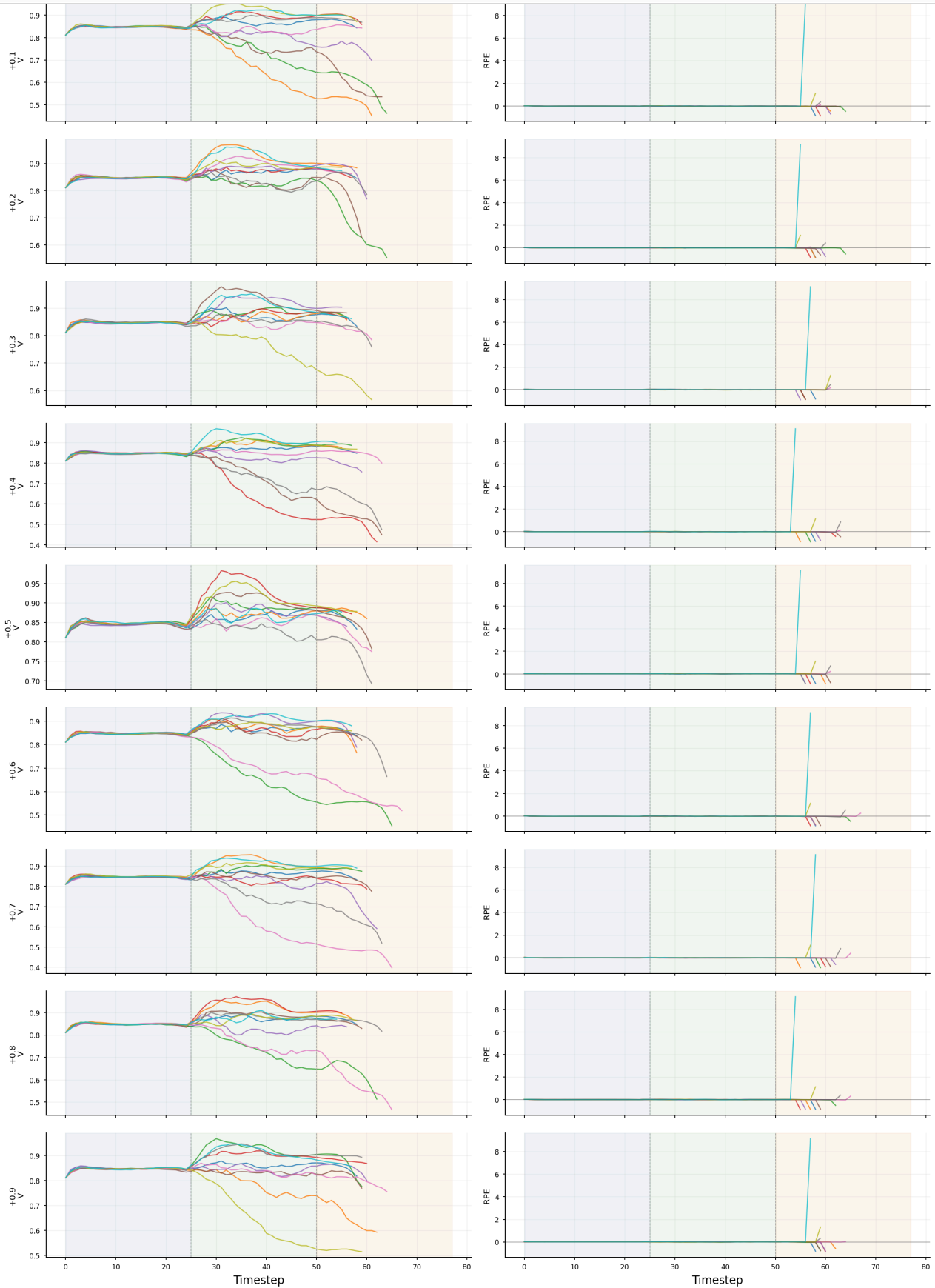
Existing: V and RPE dynamics across all ctx levels

Value prediction and objective RPE traces for 10 representative trials at each context level.

Vanilla OpAL04: V and TD-RPE dynamics across direct context

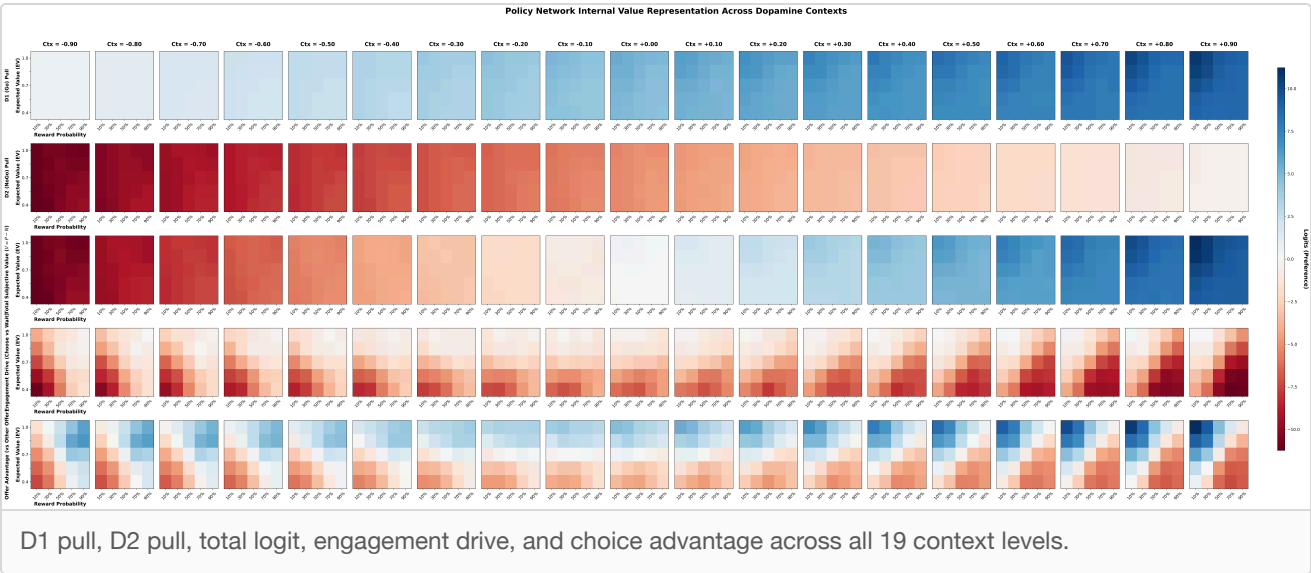


Value prediction and objective RPE traces for 10 representative trials at each context level.



Existing: Policy subjective values mega (5 rows × 19 ctx)

D1 pull, D2 pull, total logit, engagement drive, and choice advantage across all 19 context levels.



Existing: Mega comparison context sweep

Comprehensive multi-row comparison of behaviour, neural regression, and value grids across all dopamine context levels.

